

- ZERIN LABS -

Shader Pack

Cartoon sky

Welcome!

...and thanks for buying this outstanding shader pack :)

In this small tutorial you will find all the necessary details to configure the shaders and prepare your scene to create amazing backgrounds and skyboxes.

For any doubt feel free to contact us at: zerinlabs@gmail.com

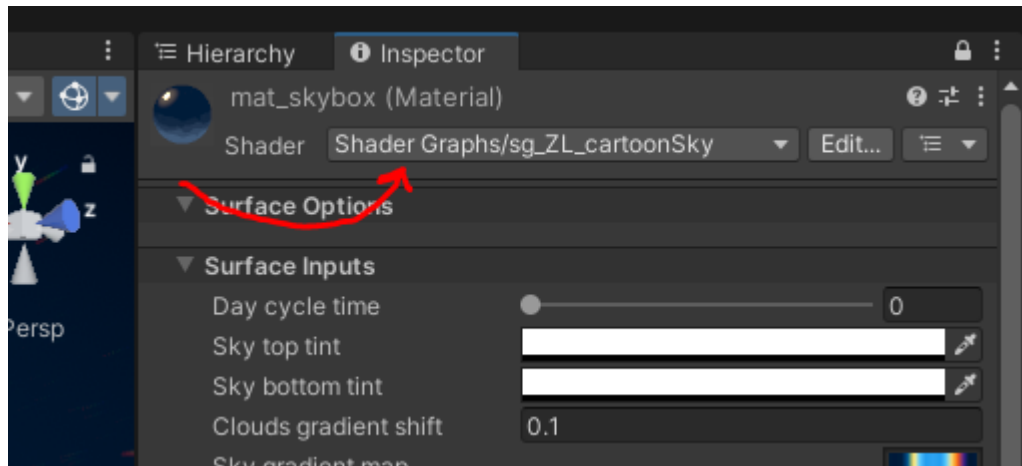
HOW TO USE THE SHADER INSIDE UNITY	2
SHADERS & PROPERTIES	3
Introduction	4
Properties	5
SKYBOX MANIPULATOR UTILITY	8
Introduction	8
How to use the manipulator	9
CONTACT	10

HOW TO USE THE SHADER INSIDE UNITY

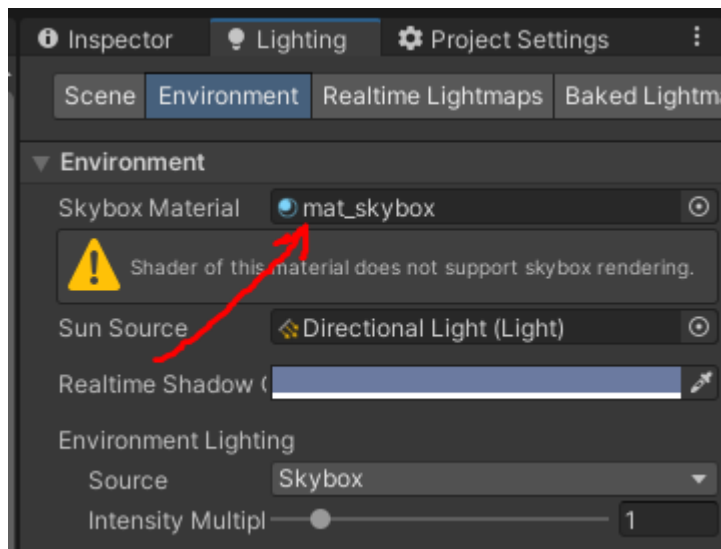
IMPORTANT: It is highly recommended you install **Shader Graph** in case you would like to tweak or modify the shaders

To use the shader inside unity you should assign it to a new skybox material.

You can do that by navigating the material shader menu and selecting the [Shader Graphs/sg_ZL_cartoonSky] shader

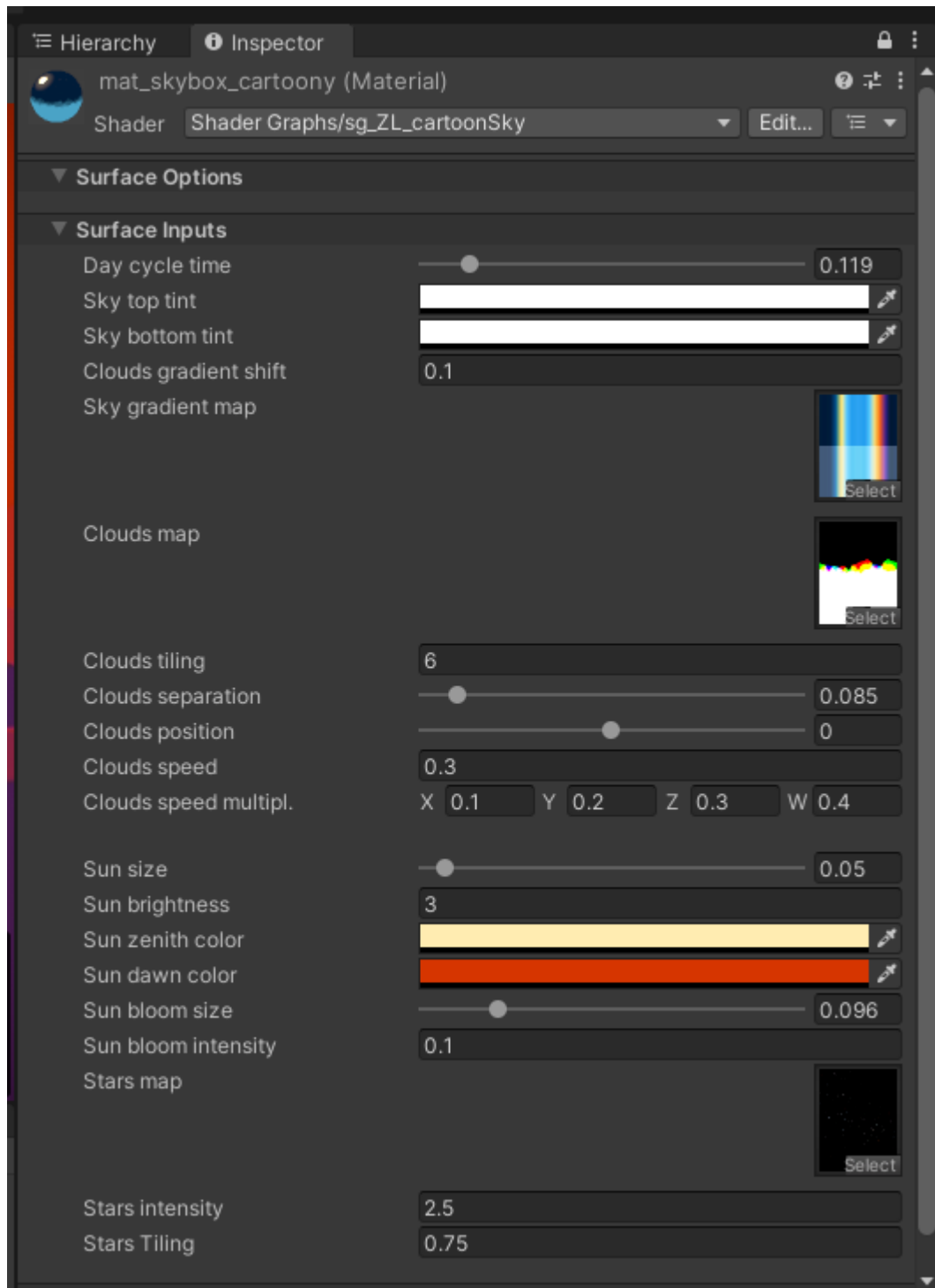


Then, navigate to unity Lighting/Environment options and assign your new material to the skybox material



SHADERS & PROPERTIES

Shader : sg_ZL_cartoonSky shader



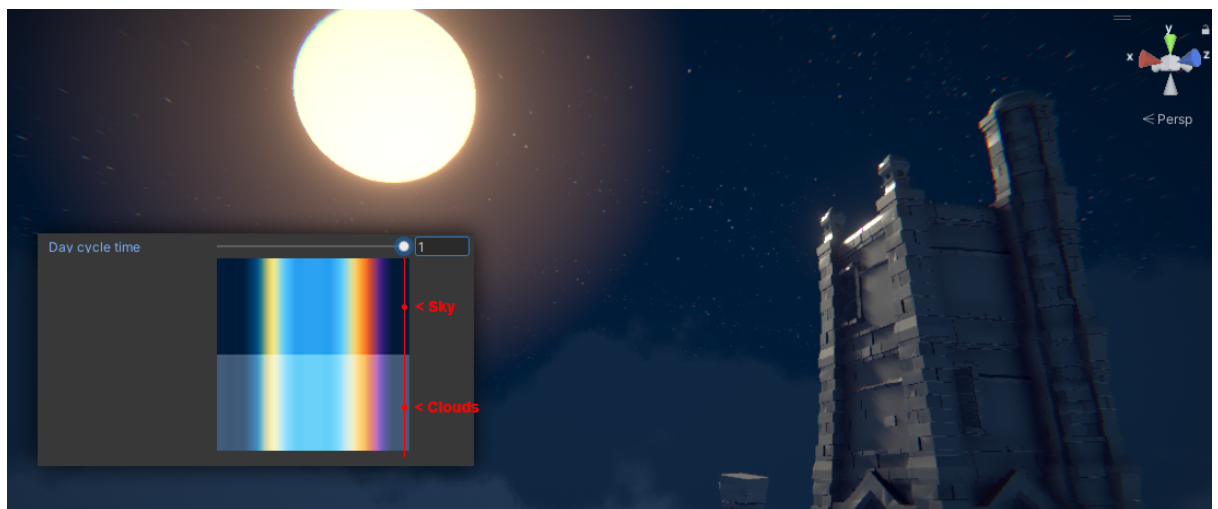
Introduction

This shader is designed to work in two different ways: as Static skybox or as a Dynamic one (controlled by script)

For a static color setup you can tweak the shader parameters at will, but it is recommended you select the cloud and sky colors using the [Sky top/bottom tint] colors as it will make your tuning easier

For more advanced users or Dynamic skies, you can use the [**Sky gradient map**] texture. This texture is a “look up table” where the sky and cloud colors will be picked.

The sky color is taken from the “upper” part of the texture, while the clouds colors are taken from the lower part.



The clouds are generated through a special 4 channel [**Clouds map**] texture. In this map the RGBA channels are used to store up to 4 different images/layers of clouds. You can create this kind of texture easily with your favorite bitmap editor package (f.i. photoshop). Simply remember to store each black and white cloud mask image on each channel respectively (R, G, B)

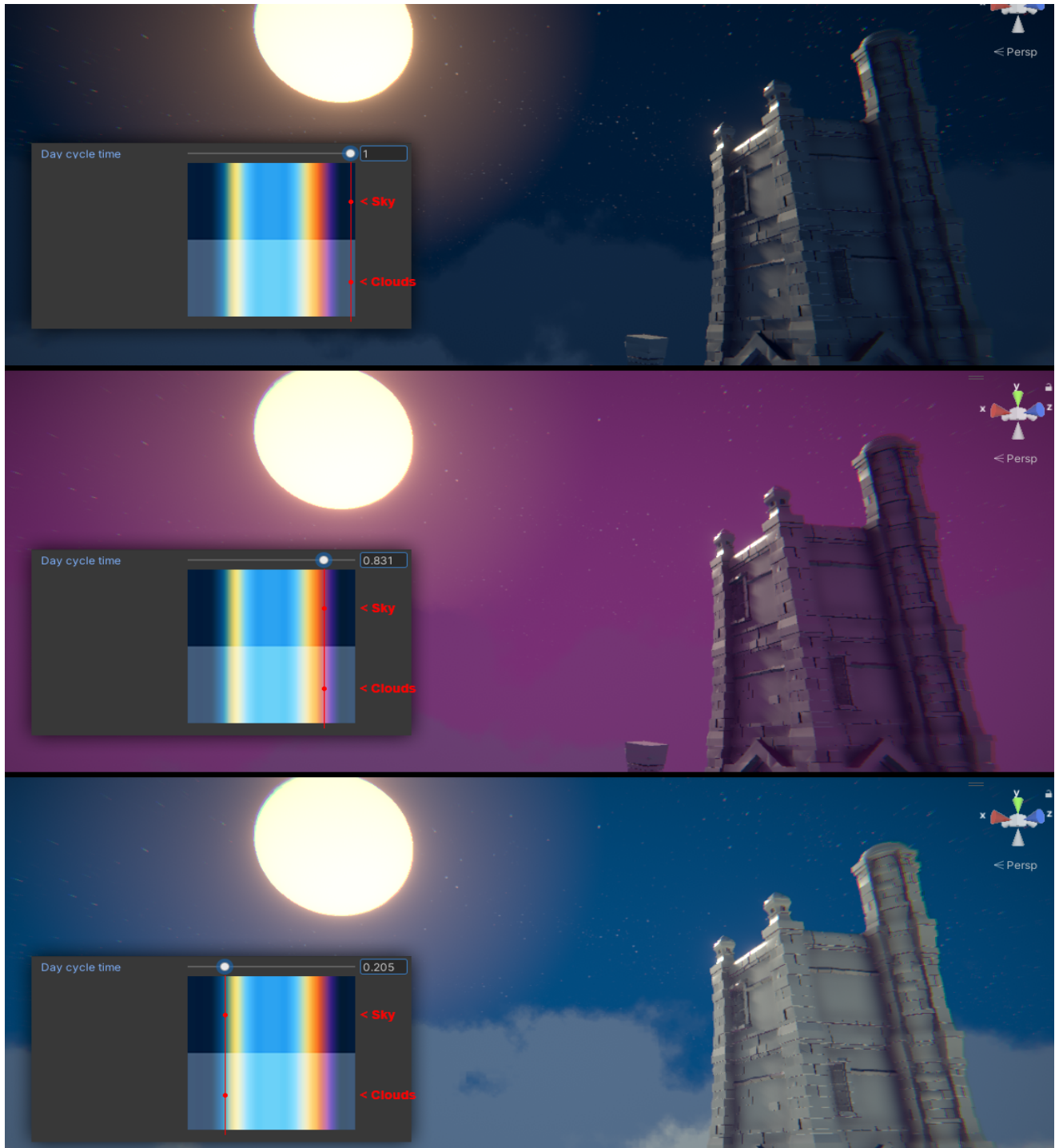
Take a look to the included textures for more references

Finally there is one more texture used for the stars field, the [**Stars map**] texture. This texture is **overlaid additively** over the sky so it is recommended to use a black (or really dark) background for it in order to avoid affecting the sky color too much.

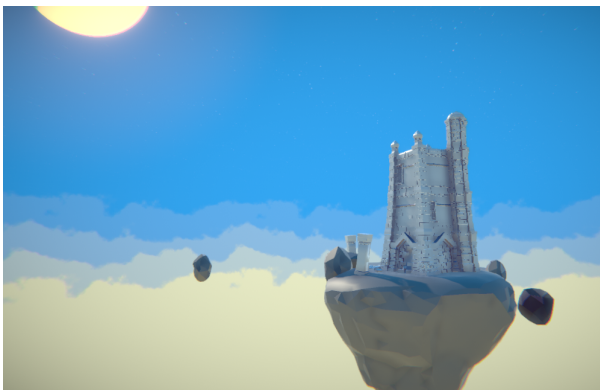
Properties

- **Day cycle time**

- This parameter allows to “scroll” the [Sky gradient map] texture in order to color the sky and clouds dynamically.
- **IMPORTANT:** You can animate this parameter (and the rest of parameters as well) using the [scr_skyboxManipulatorUtility] script utility included with this pack.

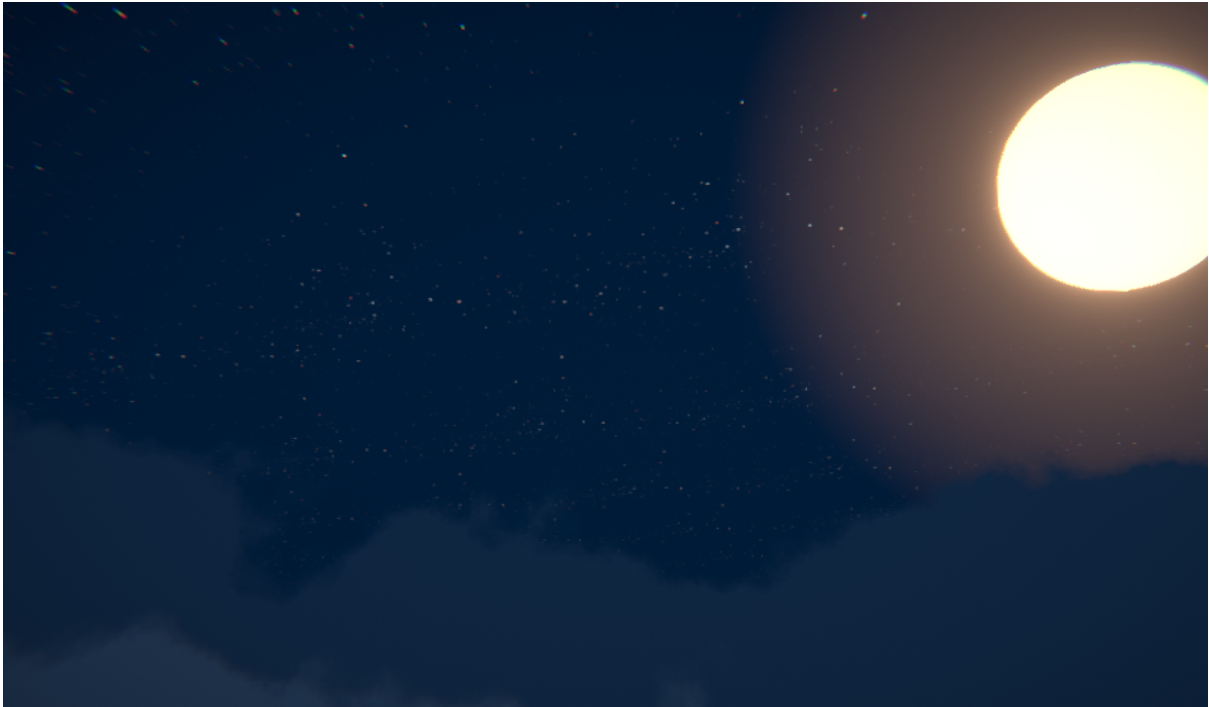


- **Sky top tint**
 - This color is used to tint (multiply) the sky color.
 - If you're not using the [**Sky gradient texture**] this color will be the general "sky color" and the clouds will transition from this color to the [**Sky bottom tint color**]
- **Sky bottom tint**
 - This color is used to tint (multiply) the clouds color.
 - Clouds color transition from the [**Sky top tint**] color to this if used standalone.
- **Clouds gradient shift**
 - If you're using the [**Sky gradient map**] this parameter will shift a bit the pointer of the look up table in order to make a bigger difference between the sky and the clouds colors
- **Sky gradient map**
 - Texture used as a look up table as it is explained on the introduction and the [**day cycle time**] parameter.
- **Clouds map**
 - 4 channel RGBA texture where each channel is used to store a different black and white cloud texture mask (alpha included)
 - RGB and A channels correspond to each one of the layers from background to foreground
- **Clouds tiling**
 - controls the tiling of the cloud layers



- **Clouds separation**
 - Separates the 4 cloud layers towards the top and the bottom of the skybox respectively
 - Changing this value can create very interesting parallax effects
- **Clouds position**
 - Pushes the cloud layers to the top or bottom of the sky respectively

- Changing this value can does the effect that we're above or under (or inside) the clouds
- **Clouds speed**
 - Changes the speed of the parallax of the clouds
- **Clouds speed multipl.**
 - Factor that multiplies each one of the "cloud layers" speed so they can move at different speeds
 - RGB and A channels correspond to each one of the layers from background to foreground
 - If you want to keep one of the layers "fixed" simply set the channel value to "0"
- **Sun size**
 - Changes the side of the sun circle
 - It could be used to emulate the moon too.
- **Sun brightness**
 - Changes the intensity of the sun. Changing this parameter will affect the intensity of the GI light too
- **Sun zenith color**
 - This is the color of the sun (or moon) when it is on the highest part of the sky
- **Sun dawn color**
 - The color of the sun (or moon) when it is down the horizon
- **Sun bloom size**
 - Size of the glowing halo around the sun circle
 - The sun bloom has the same color as the sun
- **Sun bloom intensity**
 - Controls the intensity of the sun glow
- **Stars map**
 - Texture used as an additive overlay over the sky
 - A dark or black background texture is recomended



- **Stars intensity**
 - Multiplier that modulates the shininess of the stars
 - Set it to zero if you want to remove the stars (for instance during day)
- **Stars tiling**
 - Controls the tiling of the stars map on the sky

SKYBOX MANIPULATOR UTILITY

Introduction

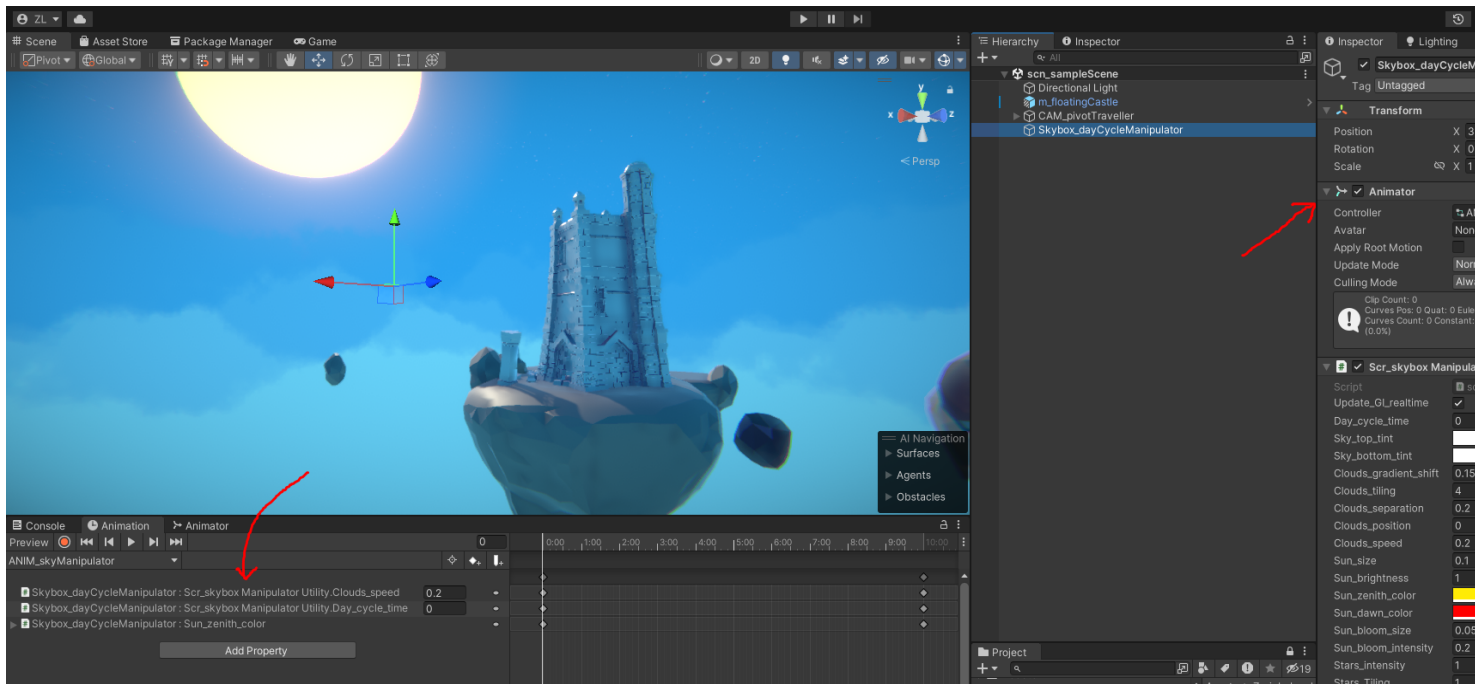
IMPORTANT: The [`scr_skyboxManipulatorUtility`] script utility is meant for advanced users that would like to control some of the skybox parameters using C# scripting.

IMPORTANT: This utility is on **BETA** stage and could produce some undesired results. It is **highly recommended** that before using this script you **clone or save a preset of your skybox material** as it can be changed unintentionally.

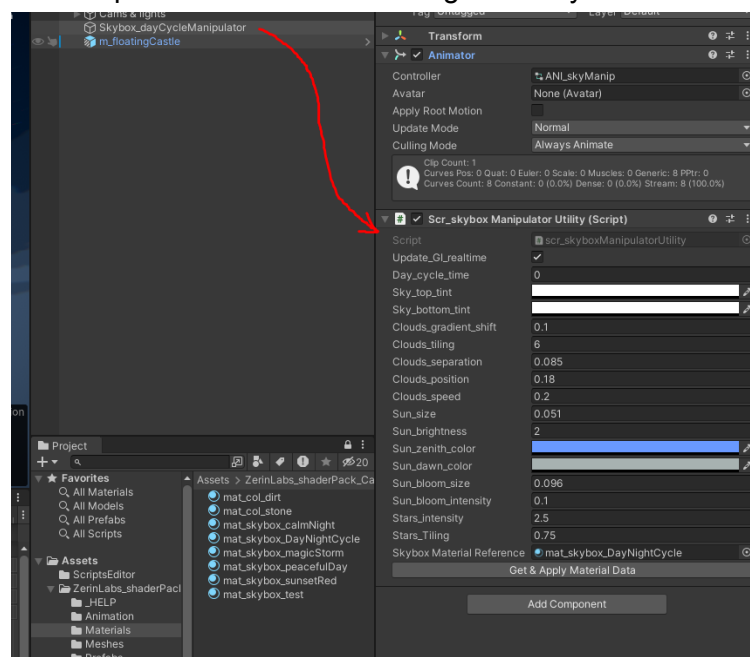
There are few properties that cannot be tweaked dynamically through this utility. This properties are: `_Sky_gradient_map` / `_Clouds_map` / `_Stars_map`

How to use the manipulator

1. Create a new Material using the [sg_ZL_cartoonSky] shader and assign the material to the unity Skybox
2. Add the [scr_skyboxManipulatorUtility] script to an empty GameObject in your scene
3. Assign an animator to your this and animate the exposed script parameters using timeline



4. The script replicate almost all the parameters of the shader so you can modify the parameters in the script and animate them using the Unity Timeline animation system



5. Once you press PLAY, the skybox will be animated accordingly

TRICK: You can use the “skybox material reference” slot to load a skybox material and set up all the parameters with the “Get & Apply Material Data” button.

You can, for instance, set-up several materials capturing different “day moments” of the skybox and use this slot+button to paste all their information into timeline keyframes for animation purposes.

TRICK: Assign the same material to a 3d sphere so you can preview the material first

CONTACT

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